

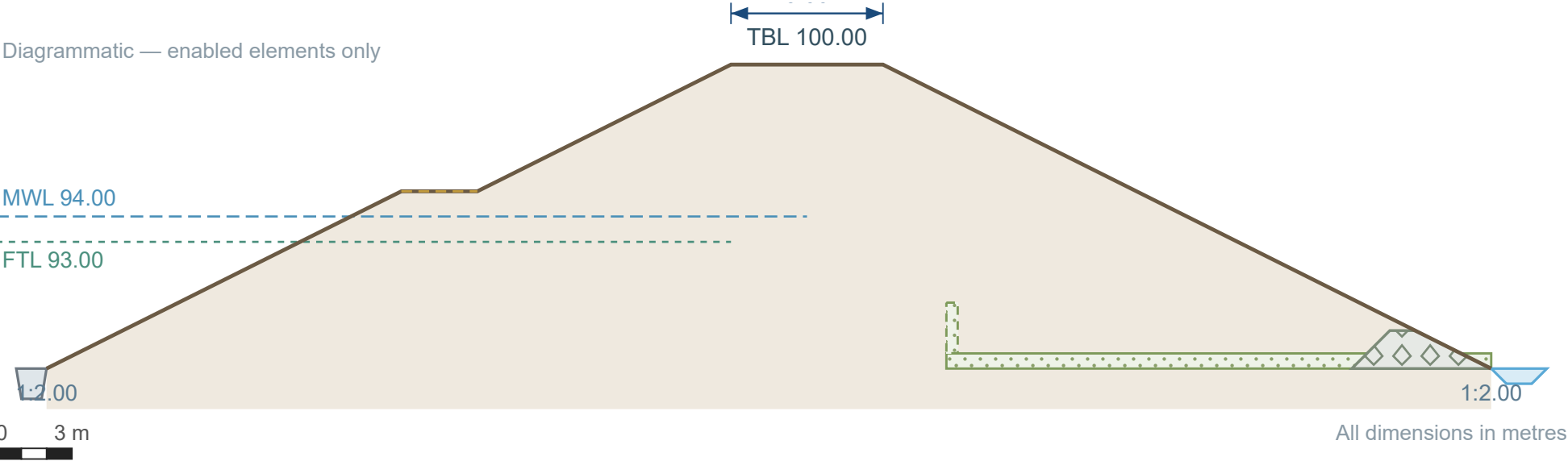
Bund #1 [Audit]

NEW · HOMOGENEOUS

Audit reservoir · 3 surveyed section(s) · Bund length 60 m · Datum 0 m · Crest width 6 m

MWL (m)	Freeboard (m)	TBL (m)	FTL (m)	U/S slope	D/S slope
94	2	100	93	2:1	2:1

Diagrammatic — enabled elements only



Statement of quantities

Average section value = (A1 + A2) / 2. Interval quantity = average section value × length.

Chainage	Length (m)	Average toe RL (m)	Bund height (m)	Bund width / surveyed ground perimeter (m)	Jungle clearance			Bund foundation excavation — depth 0.3 m			Homogeneous bund filling (excluding berms)		
					Section	Average	Total	Section	Average	Total	Section	Average	Total
					(m)	(m)	(m²)	(m²)	(m²)	(m³)	(m²)	(m²)	(m³)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
0	—	88	12.3	57	57	—	—	17.1	—	—	360	—	—
30	30	88	12.3	57	57	57	1710	17.1	17.1	513	360	360	10800
60	30	88	12.3	57	57	57	1710	17.1	17.1	513	360	360	10800
Total	60						3420			1026			21600

Statement of quantities — continued

Chainage	Length (m)	U/S toe-wall excavation			D/S toe-drain excavation			Rock toe			Rock-toe foundation excavation		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(15)	(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)	(26)
0	—	1.2	—	—	0.78	—	—	4.875	—	—	0	—	—
30	30	1.2	1.2	36	0.78	0.78	23.4	4.875	4.875	146.25	0	0	0
60	30	1.2	1.2	36	0.78	0.78	23.4	4.875	4.875	146.25	0	0	0
Total	60			72			46.8			292.5			0

Statement of quantities — continued

Chainage	Length (m)	Rock-toe graded filter			Horizontal filter			Vertical filter			Chute-drain excavation		
		Section	Average	Total	Section	Average	Total	Section	Average	Total	Section	Average	Total
		(m ²)	(m ²)	(m ³)	(m ² / m)	(m ² / m)	(m ³ / m ²)	(m ² / m)	(m ² / m)	(m ³ / m ²)	(m ²)	(m ²)	(m ³)
(1)	(2)	(27)	(28)	(29)	(30)	(31)	(32)	(33)	(34)	(35)	(36)	(37)	(38)
0	—	6.478	—	—	12.9	—	—	0.9	—	—	0.161	—	—
30	30	6.478	6.478	194.34	12.9	12.9	387	0.9	0.9	27	0.161	0.161	4.83
60	30	6.478	6.478	194.34	12.9	12.9	387	0.9	0.9	27	0.161	0.161	4.83
Total	60			388.68			774			54			9.66

Statement of quantities — continued

Chainage	Length (m)	Chute-drain lining			Berm filling (U/S, RL 95)		
		Section	Average	Total	Section	Average	Total
		(m ³ /m)	(m ³ /m)	(m ³)	(m ²)	(m ²)	(m ³)
(1)	(2)	(39)	(40)	(41)	(42)	(43)	(44)
0	—	0.107	—	—	21	—	—
30	30	0.107	0.107	3.22	21	21	630
60	30	0.107	0.107	3.22	21	21	630
Total	60			6.44			1260

Earthwork excavation classification

Percentages are the selected dashboard classification applied to each measured source.

Foundation / stripping · 1026 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$1026 \times 100 / 100$	1026

Upstream toe wall · 72 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-DAW-1-1	$72 \times 100 / 100$	72

Downstream toe drain · 46.8 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$46.8 \times 100 / 100$	46.8

Chute drain · 9.66 m³

Soil / rock	Share (%)	Code	Calculation	Quantity (m ³)
All Soils	100	IRR-CAW-1-1	$9.66 \times 100 / 100$	9.66

Payable quantities of Excavation.

Each excavation DATA code is shown once, followed by the works contributing to its total.

IRR-CAW-1-1.

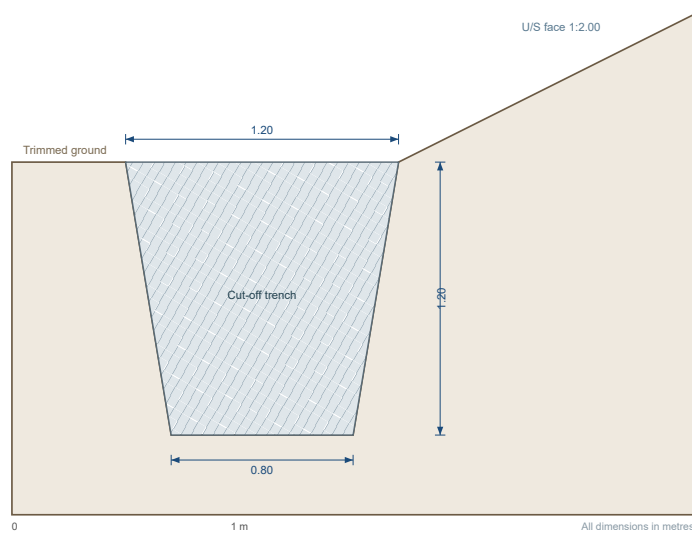
Foundation stripping 1026	Downstream toe wall + 46.8	Chute drains + 9.66	Total = <u>1082.46</u> m ³
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IRR-DAW-1-1.

Upstream toe wall 72	Total = <u>72</u> m ³
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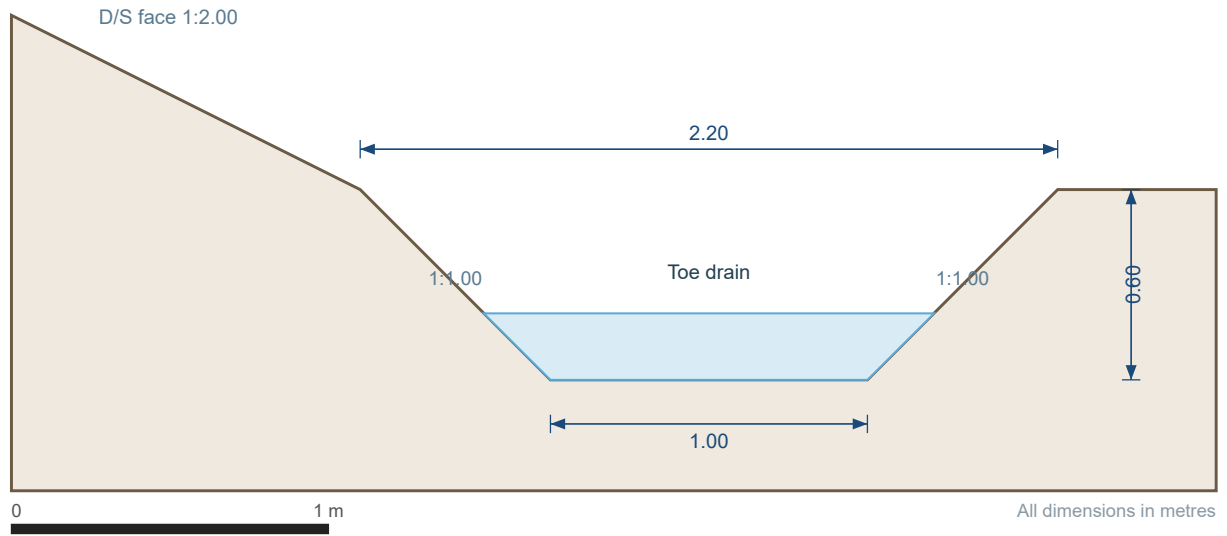
Component details

Upstream toe wall / anchorage



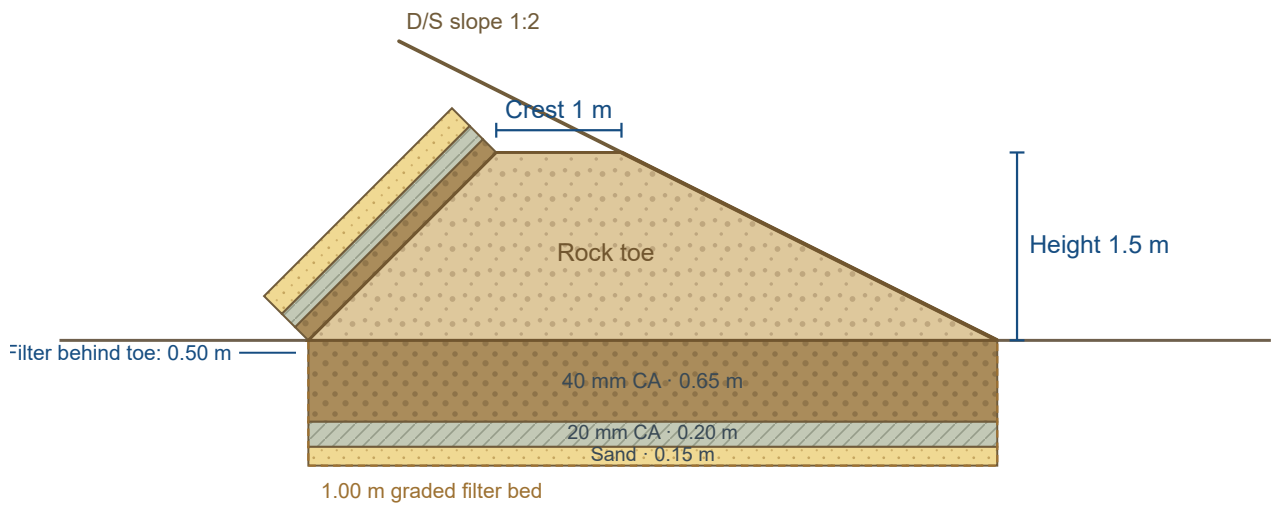
Dimension	Value
Top width (m)	1.2
Bottom width (m)	0.8
Depth (m)	1.2
Side slope (H:V)	1

Downstream toe drain



Rock toe and graded filter

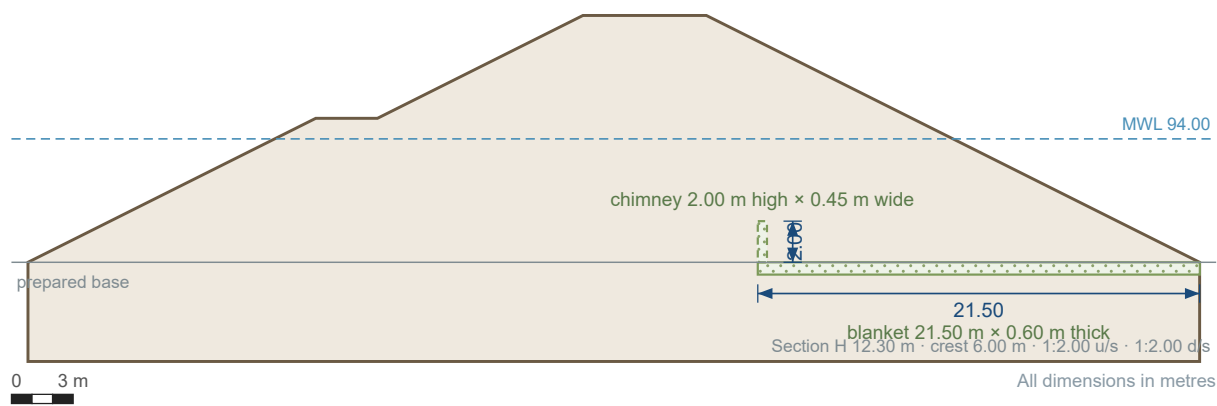
The shared general excavation beneath the rock-toe foundation is assigned to the rock-toe trench and deducted from the general cut. It is counted once.



$$\text{Area} = H \times [\text{Crest} + H \times (\text{inner} + \text{outer}) / 2] = 1.5 \times [1 + 1.5 \times (1 + 2) / 2] = 4.88 \text{ m}^2$$

Base 5.5 m · inner 1:1 · outer = D/S slope

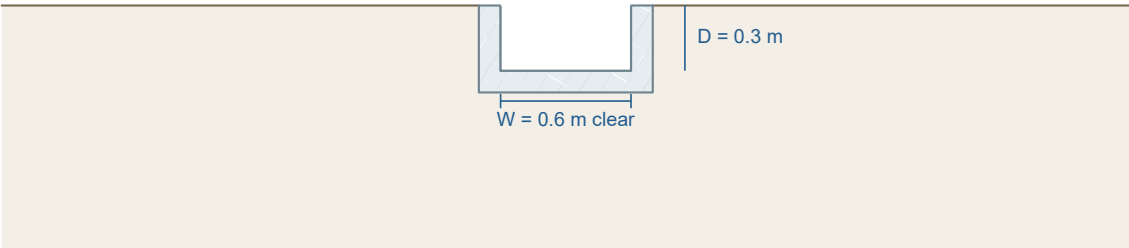
Internal drainage arrangement



Chute drains

Excavation area: $A = W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$. Lined perimeter: $P = W + 2D = 1.2 \text{ m}$. Per chute, excavation = $A \times \text{developed length}$; protection = $P \times \text{developed length} \times \text{lining thickness}$.

CC protection: $t = 0.1 \text{ m}$



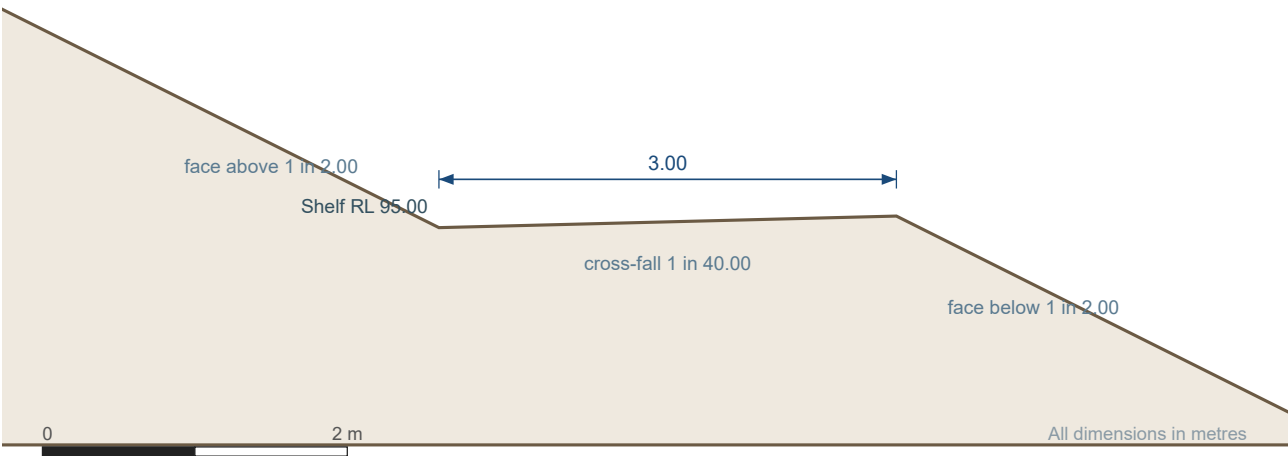
Excavation area = $W \times D = 0.6 \times 0.3 = 0.18 \text{ m}^2$
Lined perimeter $P = W + 2D = 0.6 + 2(0.3) = 1.2 \text{ m}$

Cross-section of downstream chute drain

Berm shelf and catch-water drain

The shelf is included in the embankment geometry; only its selected surfacing and drain works are additional quantities.

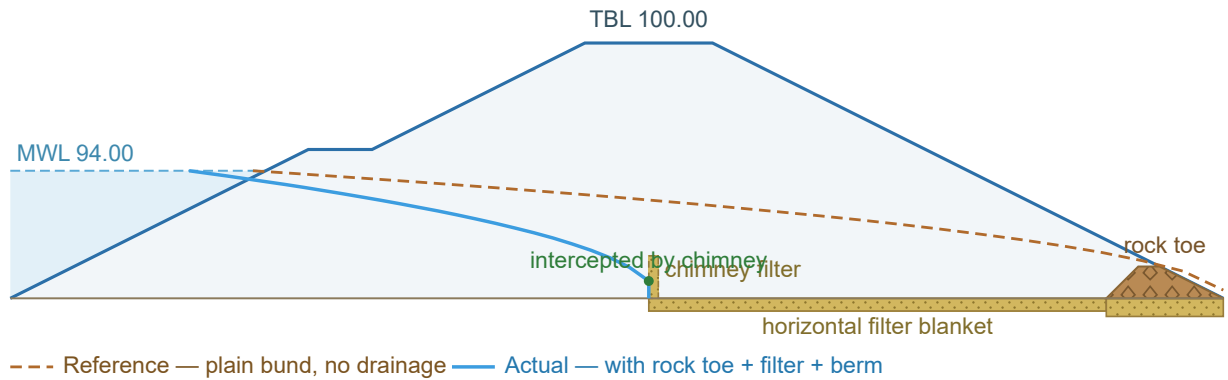
shelf is part of the design face



Phreatic-line comparison

Reference and selected drainage conditions at the governing cross-section.

Parameter	Without drainage	Selected drainage
Focus datum RL (m)	88	88
Water depth (m)	6	6
Focal distance S; $q = K \times S$ (m)	0.393	0.818



Payable quantities

Selected work codes and quantities from the same engine as the component dashboard. Distinct operations can share a measured volume; their quantities must not be added to obtain geometric fill.

No.	Code	Description	Unit	Quantity
1	IRR-PMW-1-2		m ²	3420
2	IRR-CAW-1-1		m ³	1082.46
3	IRR-DAW-5-6		m ³	22860
4	IRR-DAW-1-1		m ³	72
5	IRR-CAW-6-1		CUM	6.44
6	IRR-CAW-5-6		CUM	292.5
7	IRR-CAW-5-5		CUM	1162.68
8	IRR-DAW-6-8		CUM	54